Disease Exposure, Environmental Health, Demographic Change, And Labor-Market Adjustment

Incidence, Reallocation, Migration, Aging, And Worker Welfare

Special Topic 6: Labor Market, Health, and Population – Week 5

Central question

- How do disease, environmental exposure, and demographic structure reshape workers, firms, local labor markets, and aggregate adjustment?
- The labor objects are employment, hours, wages, productivity, migration, retirement, technology adoption, job quality, and welfare.
- Week 5 discipline: distinguish short-run disruption, medium-run reallocation, and long-run distributional or institutional effects.

Disease, environment, and demography as labor shocks

Shock	Immediate labor margin	Adjustment margin
Infectious disease	absences, closures, risk, care	sectoral recovery, safety, migration
Pollution / heat	fatigue, productivity, hours	shifts, task redesign, sorting
Mortality crisis	scarcity, household disruption	wages, institutions, skill premia
Aging / fertility decline	participation, retirement, care	automation, migration, job design

- Incidence can appear in wages, hours, output, rents, firm exit, migration, retirement, or hidden welfare losses.

A labor-market adjustment system

$$Y_{rgt} = f(H_{rgt}, E_{rt}, D_{rt}, F_{rt}, M_{rt}, P_{rt})$$

- H: health capacity, morbidity, mortality risk, and fatigue.
- E: disease, pollution, heat, workplace, or neighborhood exposure.
- D: age structure, fertility, dependency ratios, and composition.
- F: firm demand, production technology, scheduling, and entry or exit.
- M: migration, commuting, remote work, occupation switching, and immobility.
- P: public health, shutdowns, insurance, leave, pensions, and infrastructure.

Historical health episodes and labor outcomes

Episode	Shock object	Labor margin	Persistence question
Black Death	mortality and scarcity	wages, bargaining, institutions	inequality and labor rules
Tuberculosis	public-health capacity	cohort health, productivity, mobility	human capital and urban mortality
HIV/AIDS	morbidity and mortality	wages, employment, care, skill premia	household and local adjustment
COVID	disease plus policy shock	employment, remote work, care, sectors	reallocation and scarring

- Historical episodes teach the same labor question: who bears incidence, who adjusts, and what persists?

COVID in historical perspective

- COVID is a modern epidemic labor shock with fast data, strong policy response, sectoral concentration, remote work, and school or care disruptions.
- Like older disease shocks, it bundles health risk, labor demand, public institutions, household care, migration, and inequality.
- Employment losses can reflect infection risk, shutdown policy, customer demand, firm closure, transport risk, or family constraints.
- High-frequency payroll, transaction, mobility, and vacancy data help with timing but do not measure worker welfare by themselves.

Environmental health and hidden incidence

Exposure	Visible outcome	Hidden labor incidence
Air pollution	hours, absences, output	fatigue, errors, lower job quality
Heat	time allocation, productivity	unsafe work, shift timing, recovery
Wildfire smoke	attendance and location choice	risk, caregiving, school disruption
Workplace exposure	safety, turnover, claims	stress, sorting, compensating gaps

- Wages and employment can miss productivity loss, presenteeism, risk, and the lower real value of work.

Health inequality, migration, and adjustment

- Exposure differs by occupation, neighborhood, housing, transport, race, wealth, age, migrant status, and local institutions.
- Adjustment capacity differs by remote-work feasibility, savings, insurance, credentials, legal status, family obligations, and destination access.
- Migration is both an outcome and a selection problem: movers and stayers reveal different constraints.
- Immobility is a labor-market object when workers remain in risky jobs or places because adjustment is too costly.
- Global and development settings make informality, crowding, weak insurance, and migrant vulnerability central.

Aging and demographic change

Demographic force	Labor-market margin	Firm/place response
Aging	participation, retirement, health limits	automation, accommodation, job redesign
Fertility decline	future worker supply and care burden	recruitment, human capital, migration
Dependency structure	pensions, fiscal pressure, care demand	elder care, benefits, late-career work
Composition change	skill and age mix	wage structure, sectoral change

- Demography affects labor markets through supply, demand, care, technology, migration, and retirement margins at once.

Methods and data

- Cohort-by-place designs: historical epidemics, fetal exposure, public-health campaigns, long-run outcomes.
- Event studies: outbreaks, shutdowns, heat waves, pollution spikes, mortality shocks, firm or regional exposure.
- Exposure gradients: monitors, weather, satellite data, reporting laws, regulations, occupation risk, and policy timing.
- Panels and linked data: earnings, payroll, UI, claims, mortality, attendance, output, firms, migration, and commuting.
- Threats: survival selection, avoidance, endogenous residence, migration selection, local trends, bundled policies, and spillovers.

Research Lab design

- Primary anchor: COVID labor-market incidence and unequal employment effects by worker type and occupation.
- Challenge anchor: HIV/AIDS mortality, development, skill premia, and longer-run distributional adjustment.
- Reproduce: build a synthetic COVID-style incidence profile across occupation, exposure, care burden, migrant status, and demography.
- Diagnose: classify disease risk, policy and demand collapse, caregiving, sorting, migration, firm response, and hidden welfare incidence.
- Transfer: redesign the workflow for plague, tuberculosis, HIV/AIDS, pollution, heat, migrant vulnerability, or aging and automation.

Research design checklist

1. Name the shock: disease, environment, mortality, aging, fertility, demography, or policy.
2. Name the margin: work capacity, employment, hours, productivity, wages, migration, retirement, automation, or welfare.
3. State the timing: short-run disruption, medium-run reallocation, or long-run persistence.
4. Name the unit: worker, household, firm, occupation, region, cohort, or sector.
5. Say what remains bundled: policy, demand, avoidance, migration, survival, or general-equilibrium response.

Bridge to Week 6

- Week 5 supplies the scaled-up objects: disease, environment, demography, migration, firms, places, and welfare.
- Week 6 turns the full course into research designs with explicit labor outcomes, counterfactuals, mechanisms, and welfare objects.
- The through-line is disciplined incidence: who is exposed, who adjusts, who is constrained, and what wages or employment miss.

Takeaways

- Disease, environmental exposure, and demographic change are labor-market shocks because they alter capacity, demand, productivity, mobility, and welfare.
- Historical epidemics and COVID belong in one incidence literature when we compare timing, adjustment, institutions, and inequality.
- Health inequality is central: unequal exposure plus unequal adjustment capacity determines who bears labor-market costs.
- Strong designs name the shock, margin, timing, unit, data, threat, and welfare object.